

Element D

- We used various ideas to narrow it down like its ability to dribble, safety, how easy it is to assemble and remove, the complexity, and how versatile it is.
- We want to make sure it can actually play hockey and move, maybe even shoot the puck. Along with making it easy to understand and use.
- After the design matrix I felt that it was between the Swizzle Stick and the Four Wheeler. The D1 rookie design doesn't meet up to the requirements as much as the other two designs do. March 7th, 2024

A3	Four Wheeler										
	A	B	C	D	E	F	G	H	I	J	K
1		Dribble a Large Ball	Dribble a Small Ball	Hit a Small Ball	Hit a Large Ball	Safety x2	Easy to Assemble	Ease to Remove	Design Complexity	range of motion	total
2	Swizzle Stick	8	8	8	8	7	5	8	4	7	70
3	Four Wheeler	9	8	4	5	9	3	5	6	4	62
4	d1 rookie	2	2	3	3	8	8	8	2	3	47

- Between the Swizzle Stick and the Four wheeler Design
- The four wheeler design would work, but when it comes to hitting the ball and actually playing hockey with a full stick. The swizzle stick would work best.
- The swizzle stick makes it easy to move the hockey stick around. They would be able to dribble and shoot. The design also isn't hard to understand: along with put together and transport.

Plan of Action

- Make it on onshape and get a decent CAD model of the design. We want to make a base design on onshape to show how it looks along with how it will be put together. We are going to start prototyping, and see if we can make any alterations to the swizzle stick to improve it.
- Figure out what materials we will need. We need an idea of what materials we need, so we can have a layout of what we're going to need.
- Make a plan of how we're going to make/assemble the design. We can make instructions on how to put it together and attach it to the wheelchair.

Materials

Materials	PVC	Hockey stick	PVC connectors	glue
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We'll use the PVC to build the main structure of the invention, and use PVC connectors to get the angles we need for it. It is fused together to be secure with PVC glue. Lastly, a hockey stick to have as a reference and make sure everything is working right.

CAD Work

